AC9M6ST02 • Year 6 Mathematics

Evaluating Statistically Informed Arguments

Check data sources, samples, displays, summaries and claims

We are learning to…

Students inspect question wording, sample selection, missing context, graph scale, selected statistics and causal language before deciding how strongly data support a claim.

Represent | Reason | Calculate | Interpret

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

analyse statistically informed arguments presented in traditional and digital media; discuss and critique how data are represented and used to support claims

Audit a media statistics claim

1 **Claim**
‘9 in 10 prefer Brand A’

2 **Source**
Who conducted and funded it?

3 **Sample**
How many, selected how, from where?

4 **Question**
Was wording leading?

5 **Display**
Are scales and categories complete?

6 **Statistic**
Count, percentage, mean, mode?

7 **Conclusion**
What is actually supported?

A numerical statement can be accurate yet misleading if the sample, denominator, comparison or omitted categories are unclear.

Identify common representation tactics

tactic	risk
truncated axis	**exaggerates visual difference**
small voluntary sample	**selection bias**
percentage without denominator	**hides sample size**
average without distribution	hides spread or extremes
association phrased as cause	overstates evidence

Critique the argument, not the people. State what additional information would allow a stronger judgement.

Build and annotate the model

Represent evaluating statistically informed arguments and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Any percentage treated as strong evidence:** Check denominator and sample.
- **Graph style trusted without values:** Read axes and labels.
- **Mean considered typical automatically:** Inspect distribution and outliers.
- **Correlation described as cause:** Causal claims need stronger design and reasoning.

Quick check

- Ask who/whom/how many.
- Check denominator.
- Spot truncated axis.
- Question causal language.
- Request missing evidence.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.